Every Round Is a Release: Communication-Budgeted Federated Learning Over Constrained Links

Abraheem Rashid

***Abstract*—Cross-silo federated learning (FL) over constrained links pays for every synchronisation twice: as a transfer and, under client-level differential privacy (DP), as a release. A schedule computed from client updates is itself a release that no accountant charges, visible to anyone who sees the traffic. This article presents TierFed, which turns a byte quota into per-tier synchronisation intervals, and phases that cap the payload of any single round, while a per-site head stays private. Its programme has a closed form whose Hölder bound Ψ, measurable at the end of warm-up, limits how much any per-tier schedule can beat a uniform period at the same quota. Under a validation-selected protocol on a readmission benchmark with 20 sites, Ψ is 0.1%: TierFed reduces to the uniform period, and every federated method except AdaComm is within 0.0032 AUROC of FedPer, the multi-rate periods at 65.3 of FedPer's 217.8 MB. On a ResNet Ψ is 71.4% and TierFed differentiates the tiers as predicted, yet its gain over the best uniform period at matched bytes is −0.0020 AUROC, inside seed noise. Staggered phases cut window violations at an unchanged mean. Under client-level DP every schedule that releases is far below local training, and the traffic of an adaptive scheduler reveals a shifted site. A consistency check recomputes every release count and byte, and one command regenerates every number from the released logs.**

***Index Terms*—Federated learning, communication efficiency, network resource allocation, 6G edge intelligence, differential privacy, link emulation, clinical prediction, non-IID data.**

I. Introduction

FEDERATED learning allows several institutions to train one predictive model without exchanging records [1], [2]. In the cross-silo setting that matters for healthcare the participants are a handful of sites, each behind its own firewall and connected to the aggregation server by a link that the institution's network team controls [3], [4]. The learning job is one tenant among many on an uplink with a fixed capacity, a scheduled transfer window and an audit trail, and whether the link is an institutional WAN, a 5G/6G edge slice or a satellite backhaul, the operator's question is the same: how many bytes the job pushes per window, and whether it fits [5], [6].

A server coordinates N sites over R rounds and in each round requests some, all or none of the shared parameters. It must keep the mean fraction of the shared model transmitted per round within a quota ρ, keep the personalised accuracy, sample-weighted over sites and read at the round selected on validation, non-inferior to full-rate personalisation, and, where client-level differential privacy (DP) is required, stay within a budget (ε, δ). Three properties shape the design. Parameters do not drift apart at one rate: the layers nearest the input change slowly and the classifier head is site-specific. The quota is a network quantity that must map onto a slice allocation or a transfer window. And under client-level DP every transmitting round is a release that the accountant charges, so a schedule that depends on the private updates is itself a release.

A. Rashid is with the Department of Computer Science, Brunel University London, Uxbridge UB8 3PH, U.K. (e-mail: 2546393@brunel.ac.uk).

TABLE I
Positioning by method class (–: not addressed)

Class	Timing	Budget	Head	DP-eligible
Full-rate averaging [1], [7], [8]	every round	–	shared	yes
Partial personalisation [9]–[11]	every round	–	private	yes
Fixed local period [12], [13]	one, fixed	implicit	shared	yes
Resource-adaptive frequency [14]	one, from model	resources	shared	no
Adaptive period [15]	one, from loss	–	shared	no
Lazy upload [16]	per site, from update	–	shared	no
Layer-wise adaptive [17], [18]	per layer, from data	–	shared	no
TierFed (this article)	per tier, from dispersion	ρ, peak π	private	no
Fixed multi-rate period, TierFed-DP	per tier, planned	ρ, ε	private	yes

Prior work addresses these properties separately. Compression [19], [20] shrinks each upload but sends one every round. Local SGD with a fixed period [12], [13], [21], AdaComm [15], LAG [16] and the resource-constrained aggregation frequency of Wang *et al.* [14], the closest prior work for edge systems, send less often with one period for the whole model. Partial personalisation [9]–[11], [22] keeps a head or a representation local, partial-model training [23], [24] shrinks the trained model, and layer-wise schemes set per-layer intervals from discrepancy [17] or freeze converged layers [18]. FL over wireless and satellite links is treated through joint learning-and-communication design [6] and visibility-aware aggregation [25]. Client-level DP follows [26] with Rényi accounting [27], [28]; correlated noise [29], [30] and adaptive clipping [31] improve its utility, and record-level guarantees may suit cross-silo FL better [32], [33]. Inversion [34]–[36] and membership inference [37], [38] measure what an adversary recovers. Table I summarises the gap: no published scheduler accepts a byte quota, sets per-tier intervals from measurements with a stated condition for when they beat a uniform period, withholds the head, and is evaluated under one link model and one DP accountant.

The contributions of this article are as follows.

1) **A quota-constrained multi-rate scheduler with a closed-form test of its own value** (Section III). TierFed measures a scale-free dispersion proxy per tier and solves the integer programme that minimises the drift term $\sum_g K_g^2 \hat{\Gamma}_g$ of a blockwise local-SGD bound subject

to $\sum_g w_g/K_g \le \rho$, with the classifier head never transmitted. The relaxed programme has the closed form $K_g \propto (w_g/\Gamma_g)^{1/3}$, and a Hölder bound Ψ, measurable at the end of warm-up, limits how much any per-tier schedule can improve on a uniform period at the same quota. Phases chosen by a gcd condition cap the payload of any single round at an unchanged mean.

2) **Schedules as privacy releases** (Sections III-D and V-E). A schedule computed from client updates is not covered by the client-level Gaussian accountant and is visible in the traffic metadata; a timing experiment measures what a passive observer learns from it. The DP-eligible schedules are compared at a matched budget with full participation and against local training.
3) **An audited evaluation** (Sections II, IV and V). Rounds are selected on per-site validation splits; every baseline has its own knob swept; local, centralised and gradient-boosted references bound the task; a heterogeneity sweep and a deeper backbone test the regimes Ψ points to; and a consistency check recomputes every release count, byte and DP charge from the stated rules. Every number is regenerated from released logs by one command.

On the main benchmark Ψ is 0.1%, TierFed reduces to a uniform period, and every federated method except AdaComm reaches FedPer's validation-selected accuracy within 0.0032 AUROC.

II. System Model and Study Design

A. Federation, Tiers and Multi-Rate Aggregation

N sites hold private datasets $\mathcal{D}_k$ of size n_k, $n = \sum_k n_k$, $p_k = n_k/n$. The shared objective is $F(\theta) = \sum_k p_k F_k(\theta)$ with $F_k(\theta) = \mathbb{E}_{\xi\sim\mathcal{D}_k}[f(\theta;\xi)]$. In round r each participating site initialises the shared part of its model from the broadcast state, performs one local epoch of $s_k = \lceil n_k/B \rceil$ optimiser steps with batch size B, and returns the shared slice that is due. Parameters are partitioned into disjoint tiers $\mathcal{G} = \{\mathrm{s},\mathrm{m},\mathrm{f}\}$ (slow, medium, fast) by their position in the forward computation graph (input projection, hidden blocks, head), with the classifier head always in the fast tier. The fast tier is private, so FedPer-style personalisation is the special case $K_\mathrm{s} = K_\mathrm{m} = 1$. With $\mathcal{A}_r \subseteq \mathcal{G}$ the tiers due in round r and $\mathcal{S}_r$ the participating sites, the server aggregates the active tiers only,

$$\theta_g^{(r+1)} = \begin{cases} \sum_{k\in\mathcal{S}_r} n_k\, \theta_{k,g}^{(r)} \big/ \sum_{k\in\mathcal{S}_r} n_k, & g \in \mathcal{A}_r, \\ \theta_g^{(r)}, & g \notin \mathcal{A}_r, \end{cases} \tag{1}$$

and sites keep their own fast tier throughout. With w_g the share of the shared payload carried by tier g, round r sends the fraction $P_r = \sum_{g\in\mathcal{A}_r} w_g$ of it. The payload fraction over the R rounds and its limit after warm-up are

$$\Pi_R(\mathbf{K}) = \frac{1}{R}\sum_{r=1}^{R} P_r, \qquad \Pi_\infty(\mathbf{K}) = \sum_{g\in\mathcal{G}_\mathrm{sh}} \frac{w_g}{K_g}, \tag{2}$$

where $\mathcal{G}_\mathrm{sh} = \{\mathrm{s},\mathrm{m}\}$ and $\mathbf{K} = (K_g)_g$. Rounds with $\mathcal{A}_r = \emptyset$ transmit nothing; these local-only rounds are how a quota below $\min_g w_g$ is realised.

B. Links and Network Cost

The job is costed on a link profile (β, ℓ, P): uplink rate, round-trip latency and transmit power. With independent per-site links, a round in which site k sends b_k bits costs

$$t_r = \max_k \frac{b_k}{\beta} + \ell, \qquad E_r = \sum_k \frac{b_k}{\beta} P, \tag{3}$$

and nothing when no site sends. When the sites share one uplink (time division), $t_r = \sum_k b_k/\beta + \ell$ instead, and bytes saved convert one-to-one into time. Window compliance is the fraction of rounds whose t_r fits a transfer window W, the peak load is the largest per-site upload of any round, and the radio wake-ups of a site are the rounds in which it transmits: on cellular radios each wake-up costs a promotion delay and a high-power tail that do not depend on the payload [39], [40], so a sparse schedule saves energy beyond its bytes. Three profiles are used: cell-edge 5G NR (5 Mbit/s, 30 ms, 0.2 W) [41], an LEO satellite uplink (10 Mbit/s, 40 ms, 2 W) and a wired institutional WAN (100 Mbit/s, 10 ms). They are representative targets, not measurements. The model covers the uplink only and omits scheduling, fading and retransmission; it is a cost translation of measured bits, whose implementation Section V-D checks over paced TCP.

C. Data, Sites and Metrics

Diabetes 130-US Hospitals [42] is the main benchmark: 30-day readmission from encounter-level features. Cleaning follows Strack *et al.* (encounters discharged to death or hospice removed, first encounter per patient kept), which leaves one encounter for each of 69,990 patients, with 160 encoded features and a positive rate of 9.0%; a patient-grouped fifth (13,998) is set aside and used in no reported result, and the other 55,992 are assigned to the sites. The public release carries no hospital identifier, so sites are formed from the medical-specialty column by longest-processing-time bucketing with a client-share cap of 15% (a specialty above the cap is split, patient-grouped, before bucketing); the largest site holds 12% of the training data and the mean label total-variation distance between sites is 0.009. These clients are called *sites* rather than hospitals. Because this split is nearly IID, the same data are also partitioned by Dirichlet label skew with $\alpha \in \{0.1, 0.3, 1, 10\}$. **Fed-Heart-Disease** [43], [44], the four UCI Heart Disease collections as four real institutions with 736 training patients, and PneumoniaMNIST [45] under Dirichlet label skew appear in the supplement under the earlier protocol. The eICU-CRD [46] needs credentialed access and was not run; its open demo [47] has too few stays per hospital for a federation. Table II summarises the setup; every dataset statistic is read from the run logs.

Metrics. Each site's patients are split into training, validation and test sets (70/10/20%, patient-grouped), and every method is evaluated on each site's test split with that site's own parameters, sample-weighted; single-class splits are excluded identically for every method. The primary metric is the test AUROC at the round with the highest validation AUROC (the earliest on ties), with the bytes uploaded up to that round; test data never select anything. Secondary metrics are

TABLE II
STUDY SETUP, READ FROM THE CONFIGURATION FILE AND THE RUN LOGS

Item	Value
Main data	Diabetes 130-US: 69,990 patients, 55,992 of them at the sites, 160 features, positive rate 9.0%, 20 specialty sites (share cap 15%) or Dirichlet $\alpha \in \{0.1, 0.3, 1, 10\}$
Other data	Fed-Heart-Disease (4 institutions), PneumoniaMNIST (supplement)
Backbones	MLP 128–128–64 (45,506 parameters; tier shares 45.3/54.4/0.3%); tabular ResNet (width 128, depth 6)
Training	60 rounds, one local epoch, batch 64, Adam 10^{-3} (persistent state), inverse-frequency class weights; per-site patient-grouped split 70/10/20%
TierFed	$T_w = 5$, $K_{\max} = 8$, $\rho = 0.25$ (post-warm-up) unless stated; exact solver
Baselines	FedProx $\mu = 10^{-2}$; FedLAMA-style interval 5, factor 2; AdaComm initial period $\{4, 8, 16\}$; LAG $\xi \in \{1, 5{\cdot}10^3, 10^4\}$, sample-weighted; uniform periods $K \in \{1, 2, 4, 6, 8\}$ and staggered $(4, 4)$ on (9), $T_w = 5$
DP	$C = 1$, $\delta = 10^{-5}$, $\varepsilon^\star = 4$, $q = 1$; z calibrated per method; planner $\lambda = 1$
Seeds, tests	5 to 10 seeds (Section IV); paired Wilcoxon tests, bootstrap intervals

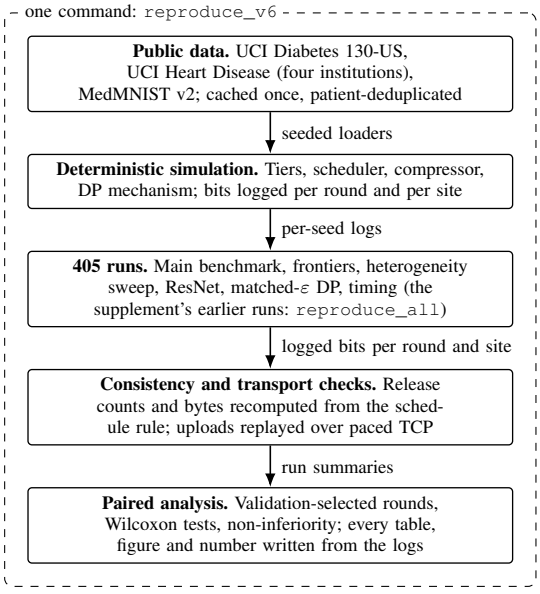


Fig. 1. Study pipeline from the public data to every reported number. All stages run from one command, and the determinism check runs first.

the mean over the last ten rounds, the mean over the post-synchronisation rounds of the last twenty, and the final round. Calibration is reported as ECE and Brier score [48], before and after Platt scaling [49] on each site's validation split, and per-site results as the worst-site AUROC. Fig. 1 summarises the study.

Problem statement. Given a quota ρ on the payload fraction (2), optionally a cap π on the payload of any one round and a budget (ε, δ), choose the intervals $\mathbf{K}$ and phases such that the validation-selected personalised AUROC is non-inferior to full-rate personalisation and is at least that of the best uniform period at the same bytes. Under DP the schedule must be fixed before training, because the accountant charges only data-independent releases. Per-site non-inferiority is reported descriptively, not tested.

III. QUOTA-CONSTRAINED MULTI-RATE SCHEDULING

The scheduler separates *what* is sent (tiers), *when* (per-tier intervals chosen under the quota), *how large* (an optional compressor) and *how private* (the DP mechanism and its accountant). Fig. 2 shows the architecture. The server-side controller observes the received slices, the sites observe only the broadcast, and the link and the privacy mechanism sit between them.

A. Dispersion Measurement

TierFed estimates the heterogeneity of each tier from the dispersion of the slices it receives. For tier g with tensors $j \in g$,

$$\hat{\Gamma}_g^{(r)} = \frac{\sum_{j \in g} \frac{1}{|\mathcal{S}_r|} \sum_{k \in \mathcal{S}_r} \left\| \theta_{k,j}^{(r)} - \bar{\theta}_j^{(r)} \right\|_2^2}{\sum_{j \in g} \left\| \bar{\theta}_j^{(r)} \right\|_2^2 + \xi_0}, \tag{4}$$

a scale-free ratio ($\xi_0 = 10^{-12}$) that makes tiers of different magnitude comparable. It exists only for tiers received in round r; otherwise the last value is carried forward. The received slices result from $m_g = (r - t_g)\, s$ local steps since the tier's last synchronisation t_g, and the standard drift lemma of local SGD [12], [50] bounds the expected numerator by a term of order $\eta^2(m_g \sigma_g^2 + m_g^2 \Gamma_g)$. $\hat{\Gamma}_g$ can therefore grow quadratically with the interval that produced it, so the controller partly measures its own decisions. The debiased proxy $\tilde{\Gamma}_g = \hat{\Gamma}_g / (r - t_g)^2$ removes the quadratic growth of the heterogeneity term; it is released as an option but not evaluated here.

B. The Programme and Its Closed Form

Bound. The analysis concerns the shared backbone with the private heads frozen; the personalisation gap this introduces is measured empirically. Assume block-separable L-smoothness (the gradient of tier g is L-Lipschitz in θ_g and does not depend on the other tiers; this ignores cross-tier coupling and is the main idealisation), unbiased stochastic gradients with per-tier variance σ_g^2, bounded per-tier heterogeneity $\frac{1}{N} \sum_k \|\nabla_g F_k(\theta) - \nabla_g F(\theta)\|^2 \le \Gamma_g$, $F \ge F^\star$ [12], [21], and the same number s of local steps at every site. With unequal s_k, plain averaging converges towards a reweighted objective [51]; the experiments keep $s_k = \lceil n_k / B \rceil$ and do not normalise.

Proposition 1 (Multi-rate local-SGD bound)**.** *With local SGD of step size η and tier g averaged every K_g rounds, there are absolute constants c_0, c_1, c_2 such that for $\eta \le c_0 / (L K_{\max} s)$,*

$$\frac{1}{R} \sum_{r=1}^{R} \mathbb{E} \left\| \nabla F(\bar{\theta}_r) \right\|^2 \le \frac{c_1 (F(\bar{\theta}_1) - F^\star)}{\eta s R} + c_1 \eta L \sum_g \frac{\sigma_g^2}{N} + c_2\, \eta^2 L^2 s^2 \sum_g K_g^2 \left(\Gamma_g + \sigma_g^2 \right). \tag{5}$$

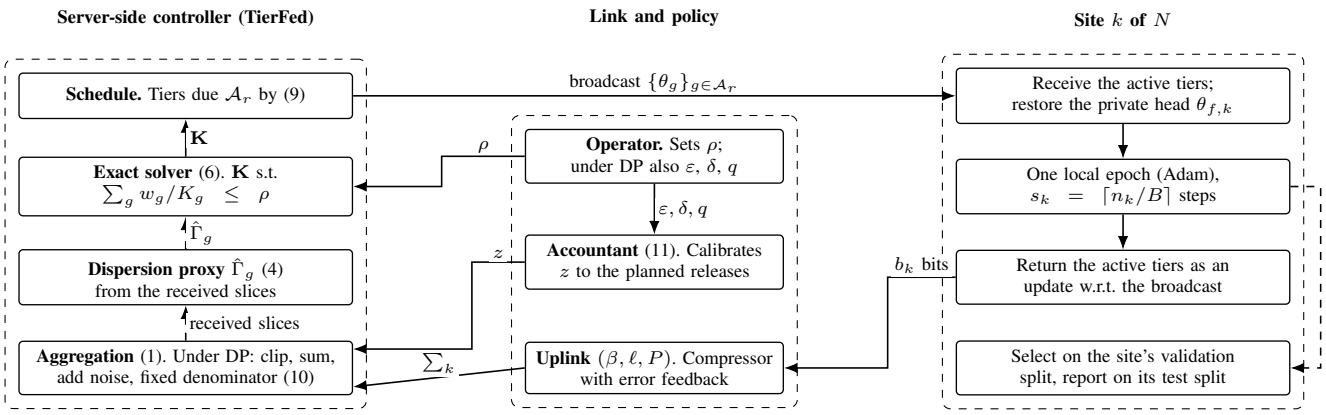


Fig. 2. Scheduler and environment. Solid arrows carry data and policy inputs; dashed arrows are measurements fed back. Under DP the dashed path from the received slices to the solver is cut and the schedule is planned before training (Section III-D).

The proof applies the perturbed-iterate argument of [21] blockwise (supplement, Section S2). In the bound, a long interval is cheap only for a tier with small $\Gamma_g + \sigma_g^2$.

Programme. Given a quota ρ the controller solves

$$\mathbf{K}^\star = \arg\min_{K_g \in \{1,\dots,K_{\max}\}} \sum_{g\in\mathcal{G}_{\rm sh}} K_g^2 \hat{\Gamma}_g \quad \text{s.t.} \quad \sum_{g\in\mathcal{G}_{\rm sh}} \frac{w_g}{K_g} \le \rho. \tag{6}$$

It keeps the $K_g^2\Gamma_g$ term of (5). The $K_g^2\sigma_g^2$ term is not separately estimable from parameter dispersion, which mixes it with the heterogeneity term, and the common factor s^2 does not change the minimiser. With the head private, $\mathcal{G}_{\rm sh} = \{\mathrm{s}, \mathrm{m}\}$ and $K_{\max} = 8$, so the programme has 64 candidates and is solved by enumeration, with ties broken towards the denser schedule.

Proposition 2 (Closed form and the value of differentiation)**.** *Let $\Gamma_g > 0$, $w_g > 0$, $\sum_g w_g = 1$, and relax K_g to the positive reals. Then (6) has the unique minimiser*

$$K_g^\star = \Big(\frac{w_g}{\Gamma_g}\Big)^{1/3} \frac{S}{\rho}, \qquad S = \sum_h w_h^{2/3}\Gamma_h^{1/3}, \tag{7}$$

with value $J^\star = S^3/\rho^2$. The uniform interval $K = 1/\rho$ costs $J_{\rm u} = \sum_g \Gamma_g/\rho^2$, and

$$\Psi = 1 - \frac{J^\star}{J_{\rm u}} = 1 - \frac{S^3}{\sum_g \Gamma_g} \in [0, 1), \tag{8}$$

with $\Psi = 0$ if and only if w_g/Γ_g is equal for all tiers. When $1/\rho$ is an integer, no integer schedule lowers the objective of (6) below that of the uniform period by more than the fraction Ψ.

The objective is strictly convex and increasing in every K_g and the feasible set is convex, so the quota binds and the KKT point is the minimiser; $S^3 \le (\sum_g w_g)^2 \sum_g \Gamma_g$ is Hölder's inequality with exponents $3/2$ and 3; the integer statement holds because the relaxation bounds the integer optimum from below and $K = 1/\rho$ is integer-feasible (supplement, Section S2). Two consequences shape the evaluation. The interval ratio $K_a/K_b = (w_a\Gamma_b/w_b\Gamma_a)^{1/3}$ compresses differences: a factor of eight in payload-to-dispersion ratio is needed for a factor of two in interval, so tiers with similar ratios collapse onto one interval after rounding. And Ψ is scale-free in Γ, so the common bias of $\hat{\Gamma}_g$ cancels when it is measured at the end of warm-up, where every tier has been synchronised every round; it then predicts, before any sparse round, how much a per-tier schedule can gain over a uniform one.

C. Schedule Rule, Phases and the Quota

Rounds are indexed $r = 1, \dots, R$. The first T_w rounds synchronise every shared tier, and thereafter tier g is due when

$$\mathcal{A}_r = \begin{cases} \mathcal{G}_{\rm sh}, & r \le T_w, \\ \{g : \ (r - T_w - \phi_g) \bmod K_g = 0\}, & r > T_w, \end{cases} \tag{9}$$

with phase $\phi_g = 0$ unless staggered. The fixed periods, the DP planner and the accountant all use this one rule, and the consistency check of Section V-A recomputes every release from it. Adaptive TierFed sends tier g when $r - t_g \ge K_g$, which coincides with (9) whenever $\mathbf{K}$ is constant after warm-up. $T_w = 5$ and $K_{\max} = 8$ are fixed across every dataset, architecture, loss and privacy setting. Fig. 3 shows one server round.

Phases and the peak. A mean quota does not bound the payload P_r of a single round: aligned tiers coincide and send the full shared payload. Two tiers with phases ϕ_a, ϕ_b are never due in the same round if and only if $\phi_a - \phi_b \not\equiv 0 \pmod{\gcd(K_a, K_b)}$ (Chinese remainder theorem), so with $\gcd \ge 2$ the offset $\lfloor \gcd/2 \rfloor$ separates them and bounds the post-warm-up peak by $\max_g w_g$ at an unchanged mean; with co-prime intervals no phase helps. The staggered variant adds $\max_{r>T_w} P_r \le \pi$ to (6) and checks it over one hyper-period $\mathrm{lcm}(\mathbf{K})$. A full-rate warm-up still sends the whole payload, so the warm-up rounds can be sent at eight bits.

The quota. Solving (6) with ρ bounds Π_∞ of (2), not the realised Π_R: a warm-up of T_w rounds at payload fraction c_w adds $T_w(c_w - \Pi_\infty)/R$ to the realised fraction. A quota over all rounds is met by solving with $\rho' = (\rho R - c_w T_w)/(R - T_w)$; both forms are reported, and the realised Π_R is read from the logs.

Compression. A compressor that emits a fraction $\kappa \in (0, 1]$ of the float32 bits multiplies the upload, $B = R\,|\mathcal{S}|\,\Pi_R(\mathbf{K})\,\kappa\,32\,d_{\rm sh}$ bits for $d_{\rm sh}$ shared parameters; it acts

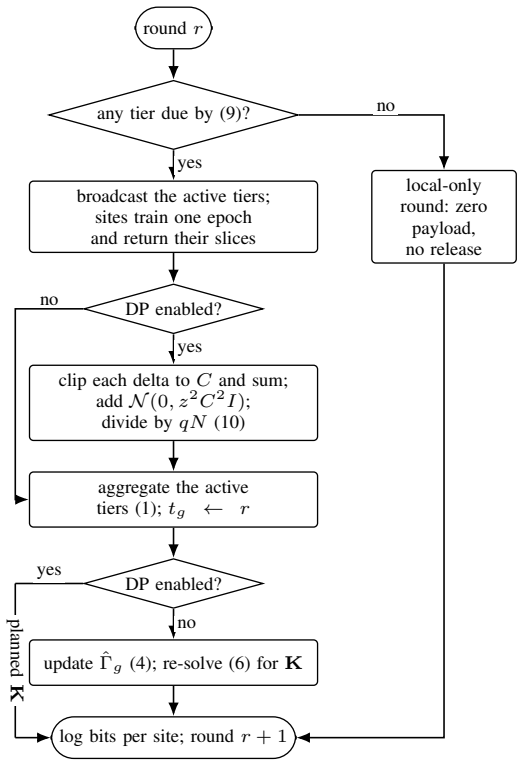


Fig. 3. One server round. Under DP the schedule is fixed before training and the dispersion feedback is never computed from the private slices.

on each site's update relative to the broadcast, with error feedback, and the bits it emits are counted.

D. Privacy: Threat Model, Mechanism and Eligible Schedules

Threat model. Sites send their updates over TLS to an aggregator that is trusted to apply the mechanism and sees raw updates, as in central DP-FedAvg [26]. Consumers of the released models, the other sites among them, see every broadcast, and central DP protects the released aggregates against them. A passive network observer cannot read the payloads but sees when each site transmits and how many bytes, which is exactly the schedule; a schedule that depends on client data therefore leaks even when every payload is encrypted. Secure aggregation with distributed noise [52] would remove the trust in the aggregator but not this metadata, and poisoning is out of scope.

Mechanism. Each round every site joins the cohort $\mathcal{S}_r$ independently with probability q; $q = 1$, full participation, is the primary cross-silo setting and $q < 1$ a sensitivity analysis. The server clips every participating site's returned delta and adds noise to the sum before dividing by the *fixed* expected cohort size,

$$\tilde{\Delta} = \frac{1}{qN}\Big(\sum_{k\in\mathcal{S}_r} \Delta_k \min\big(1, \tfrac{C}{\|\Delta_k\|_2}\big) + \mathcal{N}(0, z^2C^2I)\Big), \quad (10)$$

the fixed-denominator DP-FedAvg estimator [26], so that adding or removing one site changes the sum by at most C in ℓ_2 and the denominator reveals nothing about the realised cohort; a planned release whose cohort is empty still adds noise. The mechanism is applied by the server to every method identically.

Accountant. Over the R_{tx} planned releases the Rényi DP of the Poisson-subsampled Gaussian mechanism is composed at integer orders [28],

$$\varepsilon_\alpha = \frac{R_{\mathrm{tx}}}{\alpha-1}\log\sum_{k=0}^{\alpha}\binom{\alpha}{k}(1-q)^{\alpha-k}q^k e^{\frac{k^2-k}{2z^2}},$$
$$\varepsilon = \min_\alpha\Big\{\varepsilon_\alpha + \frac{\log(1/\delta)}{\alpha-1}\Big\}, \quad (11)$$

with $\delta = 10^{-5}$; for $q = 1$ this reduces to $R_{\mathrm{tx}}\alpha/(2z^2)$ [27]. The implementation reproduces the TensorFlow-Privacy MNIST reference ($\varepsilon \approx 3.0$) and the $q = 1$ closed form as unit tests. At $q = 1$ the exact privacy of R_{tx} Gaussian releases is μ-Gaussian DP with $\mu = \sqrt{R_{\mathrm{tx}}}/z$ [53], [54]. Both accountants depend on R_{tx} only through R_{tx}/z^2, so the exact one lowers every method's noise multiplier by the same factor, 13–24% for $\varepsilon^\star$ from 8 to 1. The comparison at matched ε is therefore unchanged, and every reported ε is an upper bound; z is calibrated to each method's own release count by bisection on (11).

Which schedules are covered. The accountant covers a release only if the decision to release does not depend on the private data. This holds for FedAvg, FedProx, FedPer, local training, a fixed period and the planner below, whose release rounds are fixed before training, and not for adaptive TierFed, AdaComm, LAG or FedLAMA, whose schedules are functions of client updates or losses; the released code refuses to run them under DP. Data-dependent release decisions are a known pattern, which privacy filters and odometers [55] and the sparse vector technique [56] charge for; the point here is that FL schedulers take such decisions without charging for them. DP-FTRL [29] and matrix-factorisation mechanisms [57] suit full participation but also need data-independent release rounds; they are not evaluated.

Planning under a privacy budget (TierFed-DP). A planner that respects the accountant chooses $\mathbf{K}$ before training from public quantities only, $(\varepsilon^\star, \delta)$, q, R, T_w, w_g and ρ. For each candidate it computes the release set of (9), the smallest $z(\mathbf{K})$ that keeps those releases $(\varepsilon^\star, \delta)$-DP, and minimises

$$J(\mathbf{K}) = \frac{\sum_g w_g K_g^2}{K_{\max}^2} + \lambda\,\frac{\sum_r P_r(\mathbf{K})\, z(\mathbf{K})^2}{\sum_r P_r(\mathbf{1})\, z(\mathbf{1})^2}, \quad (12)$$

the drift term of (5) under a flat heterogeneity prior plus the injected noise energy, each normalised by its worst case, with $\lambda = 1$. Fewer releases need less noise per release but drift more. Under the flat prior, $\Gamma_g \propto w_g$ and Proposition 2 gives $\Psi = 0$, so the planner can only choose equal intervals; it is evaluated as an ablation, not claimed as a contribution.

IV. Experimental Setup

Every v6 run uses the validation-selected protocol of Section II. Seven experiment families are run, each with every listed method on the same seeds. They are the main comparison (13 configurations, 10 seeds), the iso-byte frontier (each method with its own knob swept; 5 seeds for the added settings), the heterogeneity sweep (Dirichlet $\alpha \in$

$\{0.1, 0.3, 1, 10\}$, 5 seeds), the ResNet backbone (6 seeds), matched-ε DP at $q = 1$ (6 seeds), the timing side channel (5 seeds per world) and peak-aware staggering. Table II lists the hyperparameters. Results of the earlier final-round protocol, on compression, real institutions, imaging, scaling and a wide model, are in the supplement with that protocol stated.

Baselines and bounds. *Full rate:* FedAvg [1], FedProx [7] and FedPer [9], the reference for the non-inferiority tests. *Round-sparse:* AdaComm [15], LAG [16] with sample weighting and, for comparison with earlier versions, with the equal weights used there, a FedLAMA-style per-layer interval rule reimplemented from its published description [17], and fixed periods on (9) with the same private head and warm-up as TierFed. *Bounds:* local training without communication, a centralised MLP trained on the pooled training splits of all sites for the same number of passes, and gradient boosting [58] as a reference for model quality; all three are scored on the same per-site test splits with the round, epoch or iteration chosen on validation. SCAFFOLD with Adam is a poor match for its control variates, and its SGD arm is in the supplement. Every method uses one architecture table, one evaluation protocol and one bit-accounting rule, and nothing is chosen on test data.

Differential privacy. FedPer, fixed periods $K = 4$ and $K = 8$, one-shot federation (no warm-up, one release in round R, after a data-independent broadcast of the initial model) and the planner run at $\varepsilon^\star = 4$, $\delta = 10^{-5}$, $q = 1$, $C = 1$, each with z calibrated to its own release count. Local training is the $\varepsilon = 0$ reference on the same splits. The $q < 1$ sweeps over $\varepsilon^\star \in \{1, 2, 4, 8\}$ on 20 and 100 sites are in the supplement as a sensitivity analysis.

Timing side channel. One target site holds half of its labels flipped, or none, and an observer who sees only which rounds carry traffic and how many bytes each site sends tries to tell the two worlds apart; an AUC of 0.5 means that the schedule reveals nothing, and the fixed period is the control.

Statistics. All comparisons are paired across seeds. Two primary contrasts were fixed in the code before the runs, with the metric and both thresholds: TierFed against FedPer, and TierFed against the best uniform period at matched bytes, chosen on validation among the uniform periods whose upload is at most 5% above TierFed's; every other comparison is exploratory. Tests are two-sided Wilcoxon signed-rank tests and intervals are percentile bootstrap intervals over seeds. The gate first required a Holm-corrected $p < 0.05$, which 5 or 6 seeds cannot reach, so the second contrast is read from its interval: a setting counts as a gain only when the 95% interval excludes zero and the mean gain is at least 0.005 AUROC; no setting passes either rule. A difference below 0.005 is not called relevant, whatever its p. Non-inferiority to FedPer is tested one-sided, $H_0 : \mathbb{E}[d] \le -\Delta_{\mathrm{NI}}$ with $d = \mathrm{AUROC}_{\text{method}} - \mathrm{AUROC}_{\text{FedPer}}$, by the signed-rank test on $d + \Delta_{\mathrm{NI}}$ at $\Delta_{\mathrm{NI}} = 0.01$ and, as a sensitivity margin, 0.005. Repeated runs are bit-identical on one machine. On a second machine every data-independent schedule sends the same bytes, but the AUROC of a single run can move by up to 0.0893 (Section V-A), so every comparison is computed within one machine.

V. Results

Every table body and number is generated from the per-seed logs. Unless stated otherwise, values are means over seeds on Diabetes with the MLP and the specialty sites, p-values are from paired Wilcoxon tests, and tables S-n are in the supplement.

A. Protocol Checks

The consistency check recomputes, from the stated rules, the transmitting rounds of every data-independent schedule, the bits of every uncompressed round with full participation and the releases charged by the accountant, over all 2,218 logs of this article; it finds 0 mismatches in 977 schedule, 872 byte and 230 accountant checks. The 130 heterogeneity and timing runs repeated on a Windows laptop (all v6 runs used one Linux machine) keep their configurations; every data-independent schedule sends the same bytes in every round, and 29 of the 40 adaptive runs do, the rest differing by up to 2.9% because their decisions read client data. The AUROC at the selected round moves by a median of 0.0004 and at most 0.0893, at strong label skew: on a flat validation curve a small numerical difference selects another round (36 runs). The gate decision and the leak verdicts are unchanged.

B. Main Comparison

At the validation-selected round the federated methods are close (Table III). FedPer reaches 0.6385. Apart from AdaComm, no method differs from it by more than 0.0032 AUROC, below the 0.005 that this study calls relevant; the largest difference, for Fixed (4,4), staggered, has $p = 0.049$. AdaComm is the one federated method that is lower by a relevant margin, -0.0056 ($p = 0.004$), at 29.1 against 217.8 MB. Local training without communication differs by -0.0239 ($p = 0.002$), so federation helps on this task. On this benchmark TierFed chooses $\mathbf{K} = (4, 4)$ in every seed, and after warm-up its schedule is that of the fixed period $(4, 4)$ round for round, so the two rows are identical to the last digit. This is what Proposition 2 predicts: at the end of warm-up $\Psi = 0.1\%$, so no per-tier schedule can improve the drift surrogate over a uniform period by more than that fraction. TierFed is non-inferior to FedPer at $\Delta_{\mathrm{NI}} = 0.01$ ($p < 10^{-3}$) and at 0.005 ($p = 0.007$), with 65.3 against 217.8 MB over 60 rounds; AdaComm, at 29.1 MB, is non-inferior at 0.01 ($p = 0.007$) but not at 0.005 ($p = 0.615$).

The lead of the sparse schedules in earlier versions of this work is a property of the round at which they were read. Over the last ten rounds the sparse schedules are ahead (TierFed 0.6346, FedPer 0.6261), because full-rate training passes its best validation round early (round 17 on average, against 36 for TierFed) and then overfits; at the validation-selected round the lead disappears. Fine-tuning each site's final model for three local epochs lowers the AUROC of every method, by 0.0006 to 0.0169, so the late-round lead is not local adaptation either: sparse synchronisation overfits later. LAG's lead in the earlier table has a different cause. With the equal weights used there its final-round AUROC is 0.6337; with sample weighting

TABLE III
MAIN COMPARISON ON THE 20 SPECIALTY SITES (MLP, 10 SEEDS). AUROC AT THE VALIDATION-SELECTED ROUND WITH A BOOTSTRAP 95% INTERVAL; MB: TOTAL UPLOAD OVER ALL SITES IN 60 ROUNDS AND UP TO THE SELECTED ROUND; LAST 10: MEAN OVER THE LAST TEN ROUNDS; PHASE: MEAN OVER THE ROUNDS OF THE LAST TWENTY THAT FOLLOW A SYNCHRONISATION; +FT 3: FINAL MODEL AFTER THREE LOCAL EPOCHS AT EACH SITE; ECE BEFORE AND AFTER PLATT SCALING ON EACH SITE'S VALIDATION SPLIT; WORST-SITE AUROC; Δ AND p AGAINST FEDPER

Method	AUROC, val.-selected [95% CI]	MB	MB to sel.	Last 10	Phase	+FT 3	ECE raw/Platt	Worst site	Δ FedPer	p
Local only	0.6145 [0.6090, 0.6204]	0.0	0.0	0.5557	0.5564	0.5548	0.363/0.023	0.431	−0.0239	0.002
FedAvg	0.6391 [0.6341, 0.6448]	218.4	50.6	0.6230	0.6252	0.6069	0.365/0.024	0.468	+0.0006	0.064
FedProx	0.6391 [0.6348, 0.6438]	218.4	77.9	0.6283	0.6306	0.6108	0.361/0.025	0.488	+0.0006	0.432
FedPer	0.6385 [0.6337, 0.6437]	217.8	63.2	0.6261	0.6279	0.6083	0.367/0.024	0.468	–	–
FedLAMA-style	0.6396 [0.6344, 0.6457]	154.1	49.4	0.6314	0.6329	0.6119	0.363/0.024	0.476	+0.0011	0.131
AdaComm	0.6329 [0.6273, 0.6391]	29.1	27.7	0.6302	0.6300	0.6234	0.363/0.023	0.463	−0.0056	0.004
LAG, equal weights	0.6373 [0.6327, 0.6417]	218.4	116.9	0.6344	0.6350	0.6218	0.362/0.024	0.480	−0.0012	0.492
LAG	0.6395 [0.6342, 0.6458]	218.4	44.4	0.6237	0.6259	0.6079	0.366/0.024	0.470	+0.0011	0.064
Fixed (4,4)	0.6377 [0.6328, 0.6439]	65.3	45.7	0.6346	0.6396	0.6122	0.361/0.024	0.463	−0.0007	0.557
Fixed (4,4), staggered	0.6353 [0.6308, 0.6399]	67.0	40.2	0.6289	0.6327	0.6112	0.359/0.024	0.447	−0.0032	0.049
Fixed (8,8)	0.6372 [0.6317, 0.6440]	39.9	35.9	0.6280	0.6379	0.6003	0.363/0.024	0.457	−0.0013	0.275
TierFed	0.6377 [0.6327, 0.6440]	65.3	45.7	0.6346	0.6396	0.6122	0.361/0.024	0.463	−0.0007	0.557
TierFed, staggered	0.6353 [0.6309, 0.6399]	67.0	40.2	0.6289	0.6327	0.6112	0.359/0.024	0.447	−0.0032	0.049
Centralised MLP	0.6435 [0.6373, 0.6506]	–	–	0.5936	–	–	0.363/–	–	+0.0050	0.004
Gradient boosting	0.6413 [0.6348, 0.6491]	–	–	0.6272	–	–	0.019/–	–	+0.0029	0.160

it is 0.6212, level with FedAvg (0.6199), and at its default threshold it never skips an upload, so the lead came from the weighting.

AUPRC follows the same pattern (FedPer 0.177, TierFed 0.174, local training 0.155, prevalence baseline 0.09). Inverse-frequency class weighting shifts every method's scores, so raw ECE is 0.359–0.367; Platt scaling on each site's validation split lowers it to 0.023–0.025 for every method at no communication cost. The worst site's AUROC is 0.431–0.488: the sample-weighted mean hides sites at which no method is useful.

C. When Does Per-Tier Scheduling Pay?

Fig. 4(a) places every method on one accuracy–upload plane, each with its own knob swept, on the 5 seeds that all settings share. Above 40 MB the 18 configurations form one plateau between 0.6393 and 0.6462 AUROC, close to the centralised MLP (0.6435 over 10 seeds) and gradient boosting (0.6413). At a budget of 60 MB the configuration chosen on validation reaches 0.6461 for AdaComm, 0.6449 for the uniform period and 0.6410 for TierFed; at 30 MB only AdaComm (0.6399) and LAG at high thresholds (0.6210, near local training at 0.6197) fit. AdaComm with an initial period of 4 is $+0.0036$ above TierFed [$+0.0007$, $+0.0078$] at 54.6 against 65.3 MB, below the relevance threshold, and TierFed at $\rho = 0.15$ is -0.0020 [-0.0051, $+0.0009$] from the uniform period $K = 8$ while sending more. TierFed therefore lies on the frontier of this benchmark but does not extend it.

Table IV applies the second primary contrast in every setting. At $\alpha = 0.1$ only 14 to 16 of the 20 sites receive data, and 7 to 12 of them have a single-class test split, so that setting rests on few sites. Across the 4 Dirichlet settings, Ψ at the end of warm-up is at most 2.2%, TierFed differentiates the tiers in at most a fraction 0.04 of the post-warm-up rounds, and its gain over the best uniform period at matched bytes lies between -0.0113 and $+0.0000$ AUROC; no interval lies above zero. Label skew therefore does not move the tiers' payload-to-dispersion ratios apart on this model, while the backbone does. On the ResNet Ψ is 71.4%, and TierFed chooses $\mathbf{K} = (1, 5)$ right after warm-up in every seed, close to the interval ratio of 4.6 that (7) predicts, and differentiates the tiers in a fraction 0.88 of the post-warm-up rounds. Its difference from the best uniform period at matched bytes ($K = 4$) is -0.0020 AUROC [-0.0072, $+0.0020$], inside seed noise. A fixed period frozen at $(1, 5)$ after warm-up follows the same trajectory until round 52 at the earliest, after every selected round, so adaptation during training plays no part in this result: the per-tier split itself does not buy accuracy here. Fig. 4(c) shows every setting's gain against its Ψ.

D. Peak Load, Radio Wake-Ups and Link Cost

A mean quota leaves the payload of a single round untouched. Every transmitting round of the aligned $(4, 4)$ period sends the full shared payload, so on the 5-Mbit/s link 0.30 of all rounds exceed a window of 0.6 times the full-payload upload time (Fig. 4(d)). Staggering the two tiers keeps the mean and caps every post-warm-up round at 0.55 of the payload, so only the warm-up rounds exceed that window (0.08 of rounds); with the warm-up sent at eight bits the share is 0.00. The AUROC changes by -0.0024 ($p = 0.105$). The price is paid in radio wake-ups (Section II-B): each site transmits in 32 rounds instead of 18. A uniform period has one tier and cannot shape the peak this way; on this benchmark, this is what tier structure adds.

Over 60 rounds on the cell-edge link, communication takes 19 s for FedPer and 6 s for TierFed with independent links, and 350 s against 105 s when the sites share one uplink, where bytes convert one-to-one into time. Replaying the per-site uploads of the earlier runs over TCP paced at the model's rate matched the model to within 2.1% (supplement), which tests its implementation, not its realism.

E. Privacy: Matched Budgets and the Schedule as a Side Channel

Table V compares the DP-eligible schedules at the same budget with full participation. Every schedule that releases before its selected round is far below local training, which spends no budget: the best of them reaches 0.5415, against 0.6145 for

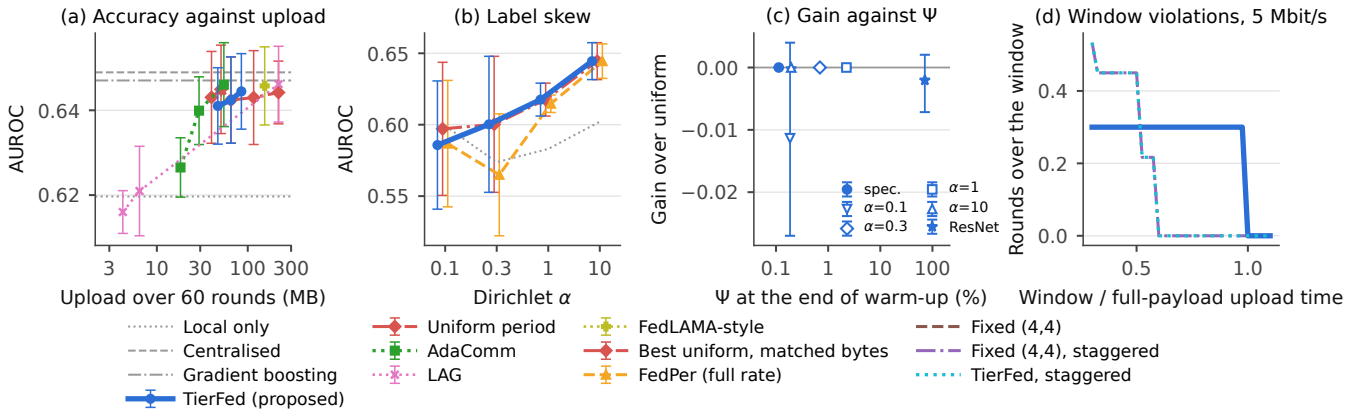


Fig. 4. (a) Test AUROC at the validation-selected round against total upload over 60 rounds, each method with its own knob swept; mean and 95% interval over seeds; grey lines: local training, centralised MLP and gradient boosting. (b) AUROC against Dirichlet label skew for TierFed, the best uniform period at matched bytes and FedPer, offset horizontally so that identical values stay visible; dotted grey: local training. (c) Gain of TierFed over the best uniform period at matched bytes against Ψ of Proposition 2 measured at the end of warm-up, one point per setting. (d) Share of rounds whose per-site upload exceeds a transfer window, against the window relative to the full-payload upload time, on the 5-Mbit/s link; identical schedules give coinciding curves.

TABLE IV

PRIMARY CONTRAST: TIERFED ($\rho = 0.25$) AGAINST THE BEST UNIFORM PERIOD AT MATCHED BYTES, CHOSEN ON VALIDATION AMONG THE UNIFORM PERIODS WITH AT MOST 5% MORE UPLOAD. Ψ: PROPOSITION 2 AT THE END OF WARM-UP; $\mathbf{K}$: TIERFED'S INTERVALS (SLOW, MEDIUM) IN THE FIRST ROUND AFTER WARM-UP, MOST FREQUENT OVER SEEDS; MB: TOTAL UPLOAD OVER 60 ROUNDS; GAIN WITH BOOTSTRAP 95% INTERVAL AND WILCOXON p

Setting	Ψ (%)	$\mathbf{K}$	TierFed	MB	Best uniform	AUROC	MB	Gain [95% CI]	p	Seeds
Specialty sites	0.1	(4,4)	0.6377	65	$K = 4$	0.6377	65	+0.0000 [+0.0000, +0.0000]	1.000	10
Dirichlet $\alpha = 0.1$	0.2	(4,4)	0.5858	52	$K = 8$	0.5971	31	−0.0113 [−0.0270, +0.0040]	0.625	5
Dirichlet $\alpha = 0.3$	0.7	(4,4)	0.6003	65	$K = 4$	0.6003	65	+0.0000 [+0.0000, +0.0000]	1.000	5
Dirichlet $\alpha = 1$	2.2	(4,4)	0.6176	65	$K = 4$	0.6176	65	+0.0000 [+0.0000, +0.0000]	1.000	5
Dirichlet $\alpha = 10$	0.2	(4,4)	0.6445	65	$K = 4$	0.6445	65	+0.0000 [+0.0000, +0.0000]	1.000	5
ResNet backbone	71.4	(1,5)	0.6413	604	$K = 4$	0.6434	602	−0.0020 [−0.0072, +0.0020]	0.844	6

TABLE V

MATCHED PRIVACY AT $\varepsilon^\star = 4$, $\delta = 10^{-5}$, $q = 1$, $C = 1$ (6 SEEDS), WITH z CALIBRATED TO EACH METHOD'S OWN RELEASE COUNT: SHARE OF UPDATES CLIPPED, AUROC AT THE VALIDATION-SELECTED ROUND, THAT ROUND, AND THE FINAL-ROUND AUROC. LOCAL TRAINING SPENDS $\varepsilon = 0$

Method	Releases	z	Clipped	AUROC	Round	Final
FedPer	60	10.04	0.44	0.5224	26	0.5205
Fixed (4,4)	18	5.50	0.39	0.5320	22	0.5272
Fixed (8,8)	11	4.30	0.38	0.5366	26	0.5303
One-shot	1	1.30	0.48	0.6199	4	0.4999
TierFed-DP	23	6.22	0.40	0.5415	32	0.5227
Local only ($\varepsilon = 0$)	0	–	–	0.6145	–	–

TABLE VI

TIMING SIDE CHANNEL (5 SEEDS PER WORLD): AUC OF AN OBSERVER WHO TELLS A TARGET SITE WITH HALF ITS LABELS FLIPPED FROM A CLEAN ONE, USING THE NUMBER OF ROUNDS WITH TRAFFIC, THE TARGET'S ROUNDS AND BYTES, OR THE BYTES OF ALL SITES; 0.5 IS CHANCE

Schedule	Traffic rounds	Target rounds	Target bytes	All bytes	Runs
Fixed (4,4)	0.50	0.50	0.50	0.50	5/5
LAG ($\xi = 2000$)	0.50	0.50	0.50	0.84	5/5
LAG ($\xi = 5000$)	1.00	1.00	1.00	1.00	5/5
TierFed	0.50	0.50	0.50	0.50	5/5

local training on the same splits. One-shot federation appears to match local training, but in every seed its selected round (round 4 on average) comes before its single release; after the release its AUROC is 0.4999. It is local training from a common initialisation, not federation. With 20 sites the noise on the average is zC/N per release, and at $\varepsilon^\star = 4$ it swamps the update whatever the schedule, while a share of 0.44 of FedPer's updates is clipped at $C = 1$. The exact accountant of Section III-D would lower z by 17% for every method alike, which does not close a gap of this size. At this federation size the result under client-level DP is therefore not a ranking of schedules but a recommendation: do not federate, or federate hundreds of sites, or protect records instead of sites.

Table VI measures what the schedule reveals to a passive observer. The fixed period sends the same bytes in both worlds by construction, and so, on this data, did TierFed, whose intervals did not change: a schedule that does not adapt reveals nothing. LAG at $\xi = 2000$ never skipped an upload of the target site, yet the other sites skipped 56% of theirs, and how often depends on the target's data through the shared model: the total upload of all sites separates the two worlds with AUC 0.84. At $\xi = 5000$ the target's own rounds give it away (AUC 1.00). Both are read from timing and volume alone, with every payload encrypted and five seeds per world. A site therefore cannot protect itself by never skipping. A scheduler cannot promise before training that it will not adapt, which is why

the accountant must refuse every adaptive schedule.

VI. Discussion

What the evidence supports. First, synchronising less costs no relevant accuracy here: every federated method except AdaComm reaches FedPer's validation-selected AUROC within 0.0032, the multi-rate periods at about a third of the upload, and the earlier lead of sparse schedules came from reading the final round. Second, whether per-tier intervals can beat a uniform period can be bounded before training: Ψ is 0.1% on the MLP, where TierFed reduces to the uniform period, and 71.4% on the ResNet, where the per-tier split chosen by the closed form still lands -0.0020 AUROC $[-0.0072, +0.0020]$ from the best uniform period at matched bytes; Ψ bounds a surrogate, and a large bound did not become accuracy. Third, phases cap the payload of any single round at an unchanged mean, which a uniform period cannot do, at the price of more radio wake-ups. Fourth, under client-level DP the schedule is part of the mechanism: a schedule that adapts to client data shows in the traffic, and at 20 sites every releasing schedule is far below local training.

What it does not support. TierFed is not more accurate than a well-chosen uniform period in any setting tested. Its quota alone is no advantage, because a uniform period meets a quota ρ with $K = \lceil 1/\rho \rceil$, and the privacy planner cannot differentiate tiers under its flat prior.

Limits of the evidence. The sites of the main benchmark are specialty buckets, not hospitals, and the four real institutions have a few hundred patients each. The link model is a cost translation of logged bits, not a network measurement (Section II-B). Proposition 1 assumes block-separable smoothness and equal local steps and is for SGD, while the benchmark uses Adam and unequal steps; Proposition 2 is exact for the programme, not for accuracy, and the stability of the dispersion feedback is observed, not proved. The privacy unit is the site: under record-level DP-SGD [32], [33] every local step is charged, so the synchronisation schedule does not change ε under standard composition. The attacks are generic; TabLeak [36] and LiRA [38] are not run. The main table and the DP comparison use 10 and 6 seeds and the other families 5 or 6; the FT-Transformer, DermaMNIST, record-level DP and site dropouts were not run under the new protocol, and on a second machine the AUROC of a single run moved by up to 0.0893.

Practical guidance. Select the reported round on validation. Measure Ψ at the end of warm-up and, when it is small, use a uniform period $K = \lceil 1/\rho \rceil$, staggering the tiers if the per-round peak matters. Under client-level DP, fix the schedule before training, calibrate z to its release count and compare with local training before federating.

VII. Conclusion

This article treated a cross-silo federated job as a network tenant with a byte quota and asked when per-tier synchronisation pays. The closed form of its programme bounds the gain of any per-tier schedule over a uniform period by Ψ, measurable before any sparse round. On the main benchmark Ψ is 0.1% and TierFed reduces to the uniform period; on a ResNet Ψ is large and the tiers are split as predicted, but the split does not beat the best uniform period at matched bytes. Phases still give tier structure something a uniform period cannot: a cap on the payload of any single round at an unchanged mean. Under client-level DP the schedule is part of the mechanism, and at 20 sites no releasing schedule comes near local training, so the quota and the privacy budget must be planned before training and checked against not federating at all.

Data, Code and Declarations

Code, configuration, per-seed logs and the script that regenerates every table, figure and number are at https://github.com/Abraheem13/tierfed. All datasets are public and de-identified; no ethics approval was required. Generative artificial-intelligence tools assisted with code refactoring, figure layout and editing; the author verified all content, code, numbers and figures. An earlier version, "Nested Federated Learning", is superseded by this article (supplement, Section S1). The author declares no competing interests and no funding.

References

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.

[2] P. Kairouz, H. B. McMahan *et al.*, "Advances and open problems in federated learning," *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.

[3] N. Rieke, J. Hancox, W. Li *et al.*, "The future of digital health with federated learning," *npj Digital Medicine*, vol. 3, p. 119, 2020.

[4] I. Dayan, H. R. Roth, A. Zhong *et al.*, "Federated learning for predicting clinical outcomes in patients with COVID-19," *Nature Medicine*, vol. 27, pp. 1735–1743, 2021.

[5] W. Y. B. Lim, N. C. Luong, D. T. Hoang, Y. Jiao, Y.-C. Liang, Q. Yang, D. Niyato, and C. Miao, "Federated learning in mobile edge networks: A comprehensive survey," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 2031–2063, 2020.

[6] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, and S. Cui, "A joint learning and communications framework for federated learning over wireless networks," *IEEE Transactions on Wireless Communications*, vol. 20, no. 1, pp. 269–283, 2021.

[7] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Machine Learning and Systems (MLSys)*, 2020, pp. 429–450.

[8] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 5132–5143.

[9] M. G. Arivazhagan, V. Aggarwal, A. K. Singh, and S. Choudhary, "Federated learning with personalization layers," *arXiv preprint arXiv:1912.00818*, 2019.

[10] L. Collins, H. Hassani, A. Mokhtari, and S. Shakkottai, "Exploiting shared representations for personalized federated learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 2089–2099.

[11] K. Pillutla, K. Malik, A.-R. Mohamed, M. Rabbat, M. Sanjabi, and L. Xiao, "Federated learning with partial model personalization," in *Proc. Int. Conf. Machine Learning (ICML)*, 2022, pp. 17 716–17 758.

[12] S. U. Stich, "Local SGD converges fast and communicates little," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.

[13] T. Lin, S. U. Stich, K. K. Patel, and M. Jaggi, "Don't use large mini-batches, use local SGD," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.

[14] S. Wang, T. Tuor, T. Salonidis, K. K. Leung, C. Makaya, T. He, and K. Chan, "Adaptive federated learning in resource constrained edge computing systems," *IEEE Journal on Selected Areas in Communications*, vol. 37, no. 6, pp. 1205–1221, 2019.

[15] J. Wang and G. Joshi, "Adaptive communication strategies to achieve the best error-runtime trade-off in local-update SGD," in *Proc. Machine Learning and Systems (MLSys)*, 2019, pp. 212–229.
[16] T. Chen, G. B. Giannakis, T. Sun, and W. Yin, "LAG: Lazily aggregated gradient for communication-efficient distributed learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018, pp. 5050–5060.
[17] S. Lee, T. Zhang, and A. S. Avestimehr, "Layer-wise adaptive model aggregation for scalable federated learning," in *Proc. AAAI Conf. Artificial Intelligence*, 2023, pp. 8491–8499.
[18] E. Malan, V. Peluso, A. Calimera, E. Macii, and P. Montuschi, "Automatic layer freezing for communication efficiency in cross-device federated learning," *IEEE Internet of Things Journal*, vol. 11, no. 4, pp. 6072–6083, 2024.
[19] D. Alistarh, D. Grubic, J. Li, R. Tomioka, and M. Vojnovic, "QSGD: Communication-efficient SGD via gradient quantization and encoding," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
[20] F. Sattler, S. Wiedemann, K.-R. Müller, and W. Samek, "Robust and communication-efficient federated learning from non-i.i.d. data," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 31, no. 9, pp. 3400–3413, 2020.
[21] J. Wang and G. Joshi, "Cooperative SGD: A unified framework for the design and analysis of local-update SGD algorithms," *Journal of Machine Learning Research*, vol. 22, no. 213, pp. 1–50, 2021.
[22] X. Yang, W. Huang, and M. Ye, "FedAS: Bridging inconsistency in personalized federated learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 11 986–11 995.
[23] E. Diao, J. Ding, and V. Tarokh, "HeteroFL: Computation and communication efficient federated learning for heterogeneous clients," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
[24] S. Horváth, S. Laskaridis, M. Almeida, I. Leontiadis, S. I. Venieris, and N. D. Lane, "FjORD: Fair and accurate federated learning under heterogeneous targets with ordered dropout," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
[25] N. Razmi, B. Matthiesen, A. Dekorsy, and P. Popovski, "Ground-assisted federated learning in LEO satellite constellations," *IEEE Wireless Communications Letters*, vol. 11, no. 4, pp. 717–721, 2022.
[26] H. B. McMahan, D. Ramage, K. Talwar, and L. Zhang, "Learning differentially private recurrent language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.
[27] I. Mironov, "Rényi differential privacy," in *Proc. IEEE Computer Security Foundations Symposium (CSF)*, 2017, pp. 263–275.
[28] I. Mironov, K. Talwar, and L. Zhang, "Rényi differential privacy of the sampled Gaussian mechanism," *arXiv preprint arXiv:1908.10530*, 2019.
[29] P. Kairouz, B. McMahan, S. Song, O. Thakkar, A. Thakurta, and Z. Xu, "Practical and private (deep) learning without sampling or shuffling," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 5213–5225.
[30] C. A. Choquette-Choo, K. Dvijotham, K. Pillutla, A. Ganesh, T. Steinke, and A. G. Thakurta, "Correlated noise provably beats independent noise for differentially private learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.
[31] G. Andrew, O. Thakkar, H. B. McMahan, and S. Ramaswamy, "Differentially private learning with adaptive clipping," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
[32] K. Z. Liu, S. Hu, Z. S. Wu, and V. Smith, "On privacy and personalization in cross-silo federated learning," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
[33] J. Liu, J. Lou, L. Xiong, J. Liu, and X. Meng, "Cross-silo federated learning with record-level personalized differential privacy," in *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2024.
[34] L. Zhu, Z. Liu, and S. Han, "Deep leakage from gradients," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
[35] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, "Inverting gradients – how easy is it to break privacy in federated learning?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
[36] M. Vero, M. Balunović, D. I. Dimitrov, and M. Vechev, "TabLeak: Tabular data leakage in federated learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023.
[37] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *Proc. IEEE Computer Security Foundations Symposium (CSF)*, 2018, pp. 268–282.
[38] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, "Membership inference attacks from first principles," in *Proc. IEEE Symp. Security and Privacy (S&P)*, 2022.
[39] J. Huang, F. Qian, A. Gerber, Z. M. Mao, S. Sen, and O. Spatscheck, "A close examination of performance and power characteristics of 4G LTE networks," in *Proc. ACM Int. Conf. Mobile Systems, Applications, and Services (MobiSys)*, 2012.
[40] A. Narayanan, X. Zhang, R. Zhu, A. Hassan *et al.*, "A variegated look at 5G in the wild: Performance, power, and QoE implications," in *Proc. ACM SIGCOMM*, 2021.
[41] 3GPP, "Study on scenarios and requirements for next generation access technologies," 3rd Generation Partnership Project, Tech. Rep. TR 38.913, 2022.
[42] B. Strack, J. P. DeShazo, C. Gennings, J. L. Olmo, S. Ventura, K. J. Cios, and J. N. Clore, "Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records," *BioMed Research International*, vol. 2014, p. 781670, 2014.
[43] R. Detrano, A. Janosi, W. Steinbrunn, M. Pfisterer, J.-J. Schmid, S. Sandhu, K. H. Guppy, S. Lee, and V. Froelicher, "International application of a new probability algorithm for the diagnosis of coronary artery disease," *The American Journal of Cardiology*, vol. 64, no. 5, pp. 304–310, 1989.
[44] J. O. du Terrail, S.-S. Ayed, E. Cyffers *et al.*, "FLamby: Datasets and benchmarks for cross-silo federated learning in realistic healthcare settings," in *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2022.
[45] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, p. 41, 2023.
[46] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, and O. Badawi, "The eICU collaborative research database, a freely available multi-center database for critical care research," *Scientific Data*, vol. 5, p. 180178, 2018.
[47] A. Johnson, T. Pollard, O. Badawi, and J. Raffa, "eICU collaborative research database demo (version 2.0.1)," PhysioNet, 2021, https://physionet.org/content/eicu-crd-demo/2.0.1/.
[48] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: the Achilles heel of predictive analytics," *BMC Medicine*, vol. 17, p. 230, 2019.
[49] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*. MIT Press, 1999, pp. 61–74.
[50] A. Khaled, K. Mishchenko, and P. Richtárik, "Tighter theory for local SGD on identical and heterogeneous data," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2020.
[51] J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, "Tackling the objective inconsistency problem in heterogeneous federated optimization," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
[52] P. Kairouz, Z. Liu, and T. Steinke, "The distributed discrete Gaussian mechanism for federated learning with secure aggregation," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021.
[53] J. Dong, A. Roth, and W. J. Su, "Gaussian differential privacy," *Journal of the Royal Statistical Society: Series B*, vol. 84, no. 1, pp. 3–37, 2022.
[54] B. Balle and Y.-X. Wang, "Improving the Gaussian mechanism for differential privacy: Analytical calibration and optimal denoising," in *Proc. Int. Conf. Machine Learning (ICML)*, 2018.
[55] R. Rogers, A. Roth, J. Ullman, and S. Vadhan, "Privacy odometers and filters: Pay-as-you-go composition," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2016, pp. 1921–1929.
[56] M. Lyu, D. Su, and N. Li, "Understanding the sparse vector technique for differential privacy," *Proc. VLDB Endowment*, vol. 10, no. 6, pp. 637–648, 2017.
[57] C. A. Choquette-Choo, A. Ganesh, R. McKenna, H. B. McMahan, J. Rush, A. G. Thakurta, and Z. Xu, "(amplified) banded matrix factorization: A unified approach to private training," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
[58] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Abraheem Rashid is with the Department of Computer Science, Brunel University London, U.K. His research interests include federated learning over constrained networks and privacy in clinical machine learning.